

logistic regression

intermediate between regression and binary classification tasks

- consider a binary classification dataset ($y^i \in \{0, 1\}$) with single feature, assume we want to estimate for each feature x^i the probability

$$p^i = P(y^i = 1) \quad p^i \in [0, 1]$$

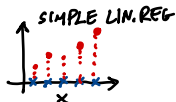
- in principle one could fit a line to data (linear regression) and set

$$p(x^i) = \hat{\beta}_0 + \hat{\beta}_1 x^i \text{ but } p(x^i) \text{ is not in the interval } [0, 1] \dots$$

OUTPUT OF A REGRESSION LINE CAN BE

$\bullet > 1$
 $\bullet < 0$

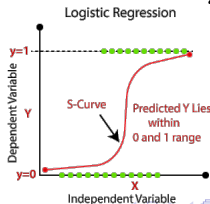
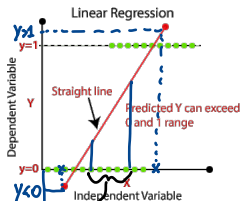
- better to fit a S-shaped nonlinear function ranging in the interval $[0, 1]$ like



LARGE RANGE OF X VALUES $\approx 0,5$

$$p(x) = \frac{1}{1 + e^{-\beta_0 - \beta_1 x}}$$

$x \rightarrow \infty \Rightarrow p(x) \rightarrow 1$
 $x \rightarrow -\infty \Rightarrow p(x) \rightarrow 0$



- logistic regression training consists in determining the parameters estimates $\hat{\beta}_0, \hat{\beta}_1$ by maximizing the **likelihood function** over TR

$p(x^i) \in [0,1] \Rightarrow$ IS A SCORE (CONFIDENCE LEVEL OF BELONGING TO CLASS 1)

$$\ell(\beta_0, \beta_1) = \prod_{i:y^i=1} p(x^i) \prod_{i:y^i=0} (1 - p(x^i))$$

HIGHLY NONLINEAR
NON CONVEX
FUNCTION
AN ITERATIVE OPTIMIZATION
ALGORITHM IS REQUIRED

- once trained the model one can use a threshold ρ (typically $\rho = 0.5$) to classify data according to

$$\tilde{y} = \begin{cases} 1 & \text{if } p(\tilde{x}) \geq \rho \\ 0 & \text{if } p(\tilde{x}) < \rho \end{cases}$$

- logistic regression easily extends to the multivariate case

$$p(x) = \frac{1}{1 + e^{-\beta_0 - \beta_1 x_1 - \beta_2 x_2 - \dots - \beta_n x_n}}$$

how to measure binary classification performance?

assume targets are {positive,negative} and the number of samples of the two classes are P and N

confusion matrix

	predicted positive	predicted negative	
actual positive	true positive (TP)	false negative (FN)	$TP+FN=P$
actual negative	false positive (FP)	true negative (TN)	$FP+TN=N$

accuracy score (ACC): $\frac{TP+TN}{P+N}$

ACC not suited for imbalanced datasets...

↓ CORRECTLY PREDICT

MANY MANY REAL-WORLD DATASETS

"a dataset is imbalanced when there is a significant disproportion among the number of samples of each class of the problem"

"trivial" example: assuming 99% of samples is $\ominus$ and only 1% is $\oplus$, a model classifying every sample as $\ominus$ achieve a 99% accuracy but is useless!

data imbalance is widespread (e.g. medical applications, spam detection,...)

alternative performance metrics can be considered to deal with imbalance

assume we are more interested in the minority class P ... $\rightarrow$ LESS REPRESENTED

PARTIAL ACCURACIES
ON EACH CLASS

confusion matrix

	predicted positive	predicted negative
actual positive	true positive (TP)	false negative (FN)
actual negative	false positive (FP)	true negative (TN)

• $TPR = \frac{TP}{P}$ (true positive rate, sensitivity, recall)

• $TNR = \frac{TN}{N}$ (true negative rate, specificity)

• $FPR = \frac{FP}{N}$ (false positive rate) $\rightarrow$ SHOULD BE HAVE BEEN CLASSIFIED AS NEGATIVE

• $FNR = \frac{FN}{P}$ (false negative rate)

• $PPV = \frac{TP}{TP+FP}$ (positive predicted value, precision)

$TP+FP \equiv$ ALL OBSERVATIONS CLASSIFIED AS POSITIVE

• $NPV = \frac{TN}{TN+FN}$ (negative predicted value)

• $BA = \frac{TPR+TNR}{2}$ (balanced accuracy)

DEPENDS ON THE CLASSIFIER

TRIVIAL EXAMPLE $\rightarrow$ B.A. $\approx 0,5$ $\rightarrow$ TOSSEING A COIN



let's focus on **precision**...

$$PPV = \frac{\# \text{ correct predicted } P}{\# \text{ total predicted } P} = \frac{TP}{TP+FP}$$

- precision is zero if all samples are predicted as $\ominus$
- precision is cheated by predicting as $\oplus$ only a small number of samples for which I am pretty sure ($FP \rightarrow 0 \Rightarrow PPV \rightarrow 100\%$)

↑ PRECISION
↑ RECALL $\Rightarrow$ THE BETTER THE MODEL

let's focus on **recall**...

$$TPR = \frac{\# \text{ correctly predicted } P}{\# \text{ total } P} = \frac{TP}{TP+FN}$$



- recall is zero if all samples are predicted as $\ominus$
- recall is cheated by predicting all samples as $\oplus$ ($FN = 0 \Rightarrow TPR = 100\%$)

cheating high precision $\Rightarrow$ bad recall (high FN)

cheating high recall $\Rightarrow$ bad precision (high FP)

F1 score: precision-recall harmonic mean (correctness-coverage trade-off)

$$F1 = 2 \frac{PPV \cdot TPR}{PPV + TPR} = \frac{TP}{TP + \frac{1}{2}(FP + FN)}$$

SUITED
FOR RATIO
MEASURES

TENDS
TO PRIVILEGE
LOWEST VALUES

RECALL

PRECISION



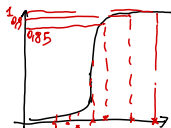
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when a threshold parameter is used to transform a classification score into binary classification (e.g. logistic regression), two issues may rise:

- the informative "probabilistic" interpretation of the classification score is ignored
- threshold is arbitrary

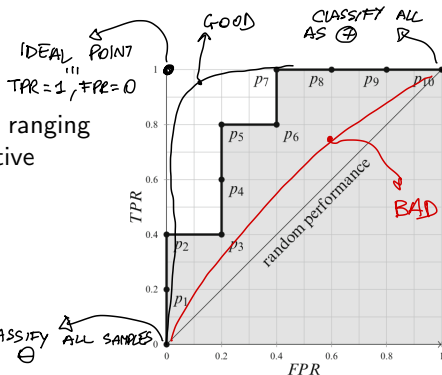
ROC (receiver operating characteristic) curve allows to evaluate such classifiers without selecting a specific threshold

- rank samples based on the score
- plot FPR and TPR for the threshold ranging from most restrictive to least restrictive



Rank	Score	Actual class	FPR	TPR	ROC point
#1	1.0	Positive	0.0	0.2	p_1
#2	0.9	Positive	0.0	0.4	p_2
#3	0.85	Negative	0.2	0.4	p_3
#4	0.7	Positive	0.2	0.6	p_4
#5	0.6	Positive	0.2	0.8	p_5
#6	0.45	Negative	0.4	0.8	p_6
#7	0.35	Positive	0.4	1.0	p_7
#8	0.3	Negative	0.6	1.0	p_8
#9	0.2	Negative	0.8	1.0	p_9
#10	0.05	Negative	1.0	1.0	p_{10}

$P=5, N=5$



AUC (area under the curve) score of a classifier is the probability to assign higher classification score to a random $\oplus$ sample with respect to a random $\ominus$ sample

exercise 4.3